A Title to the Report

A Catchy Optional Subtitle that Grabs the Attention

AB1234: Optional Course Name

Author

Delft University of Technology

TU Delft

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by

Author

Student Name

First Surname

Instructor:	I. Surname
Teaching Assistant:	I. Surname
Project Duration:	Month, Year - Month, Year
Faculty:	Faculty of Aerospace Engineering, Delft

Cover: Canadarm 2 Robotic Arm Grapples SpaceX Dragon by NASA under CC BY-NC 2.0 (Modified)

Style: TU Delft Report Style, with modifications by Daan Zwaneveld

TU Delft

Preface

A preface...

Author
Delft, April 2026

Summary

A summary...

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Nomenclature

If a nomenclature is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.

Abbreviations

Abbreviation	Definition
ISA	International Standard Atmosphere
...	

Symbols

Symbol	Definition	Unit
V	Velocity	[m/s]
...		
ρ	Density	[kg/m^3]
...		

1 Introduction

Capturing dynamic cellular interactions is a fundamentally difficult problem due to maximizing spatial and temporal resolution while minimizing sample damage. Electron microscopes can achieve sub nanometer resolution, however, that requires to first kill the sample, thus no interactions and movement cannot be observed. The optical microscope is limited by the diffraction limit. To improve upon spatial resolution and see past the diffraction limit several techniques have been developed. They all rely on fluorescent molecules, called fluorophores, which are first used to label the specime. They bind to target proteins or structures and emit light upon excitation. A sequence of frames is then captured, with molecules randomly blinking across the frames. Different fluorophores can blink either sparsely for precise localization or densely.

Super-resolution (SR) techniques can overcome this barrier. Well-known methods include Stimulated Emission Depletion (STED) microscopy **??**, Structured Illumination Microscopy (SIM) **??**, and Single Molecule Localization Microscopy (SMLM) **??**, which can achieve spatial resolution down to 10 nm. However, these methods either require complex microscope setups, high signal-to-noise ratio (SNR) conditions, or millions of frames to reconstruct a single SR image, all of which limit temporal resolution and risk sample damage **??**. Other techniques, such as Super-resolution Optical Fluctuation Imaging (SOFI) **??** and Super-Resolution Radial Fluctuations (SRRF) microscopy **??**, do not require specialized hardware and can typically operate with only hundreds of frames, accommodating dense fluorophore blinking.

Among these, SOFI is uniquely grounded in statistical theory: it reconstructs SR images by exploiting the temporal and spatial correlations of fluorophore blinking through cross-cumulant analysis. Resolution improvement arises from the statistical independence of different fluorophores **??**. Because noise is temporally uncorrelated, SOFI can operate under low SNR conditions, making it particularly well-suited for live-cell imaging. For n-th order SOFI, an n-fold improvement in spatial resolution is achieved. Despite these advantages, SOFI still requires hundreds of frames per reconstruction, which limits its ability to capture fast cellular dynamics and renders real-time imaging impractical. Without real-time capability, researchers cannot make immediate decisions during live experiments, reducing the practical utility of the technique. Deep learning offers a promising path to reduce the number of frames required for SR reconstruction while preserving resolution gains. Recent work by TODO JELLE**??** introduced a deep learning model that accelerates second-order SOFI by reconstructing a 2× SR image from as few as 20 frames. Their model encodes input frames into feature maps, fuses them with a recurrent Gated Recurrent Unit (GRU) structure to capture blinking correlations across time, and upsamples the result into a second-order SR image. Trained on synthetic data the approach demonstrated that spatiotemporal deep learning models can approximate SOFI-quality reconstructions at a fraction of the acquisition cost.

mention streaming architecture

In this thesis, we investigate whether the recurrent fusion module can be replaced by a more expressive attention-based mechanism. Specifically, we adapt TR-MISR **??** a Transformer-based Multi-Image

Super-Resolution framework originally developed for satellite imagery and ranked first on the PROBA-V benchmark, to the SOFI acceleration task. Rather than fusing frames sequentially through a GRU, TR-MISR employs a multi-head self-attention. Transformer with learnable class tokens to aggregate spatial features across all input frames simultaneously.

references

mention streaming architecture

results

1.1. Outline

In the next chapters, the concepts such as fluorescent microscopy and deep learning are presented in chapter 2. Chapter 4 presents our proposed architecture and how we augmented the datasets for moving structures followed by chapter 3. Lastly, the main outcomes are provided in chapter 3. *Introduction.. [1]*

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Background

2.1. Fluorescence Microscopy

2.1.1. Diffraction Limit

2.1.2. Point Spread Function

2.1.3. The Deconvolution Problem

2.1.4. Super-Resolution Techniques

2.1.5. Live-Cell Imaging

2.2. Super-Resolution Optical Fluctuation Imaging (SOFI)

2.2.1. Auto-cumulant

2.2.2. Cross-cumulant and Virtual Pixels

2.2.3. Flattening and Linearisation

2.3. Deep Learning for Image Super-Resolution

2.3.1. Convolutional Neural Networks

2.3.2. Recurrent Neural Networks and ConvGRU

2.3.3. Transformers and Self-Attention

Scaled Dot-Product Attention

Multi-Head Attention

Vision Transformers

TR-MISR: Transformer Multi-Image Super-Resolution

2.3.4. Autoencoders and U-Net

2.3.5. Loss Functions for Super-Resolution

Good question to flag early, examiners will notice if the background reads like a paraphrase of the prior thesis. Here's a concrete plan for each section: what to keep brief (cite Jelle), what to rewrite from a different angle, and what is genuinely new to your work.

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Experiments and Results

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3.2. Architecture Comparison: TRNet vs. SOFI-MISRGRU
3.3. Decoder Ablation: PixelShuffle vs. ConvTranspose2d
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TRNet_deconv (ConvTranspose2d decoder)
TRNet_improved (FlashAttention + CrossAttentionAggregator)
4.5. Decoder
4.6. Loss Functions
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5.2. Real Microscopy Data

5.2.1. Fixed-Cell Microtubules

5.2.2. Fixed-Cell Mitochondria

5.2.3. Data Augmentation

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Conclusion

A conclusion...